

Table creation syntax:

```
CREATE TABLE tablename(  
    Col_name Datatype,  
    Col_name Datatype  
);
```

Data Types

|

|— Numerical

| |— TINYINT → Age TINYINT (e.g., 25)

| |— SMALLINT → Year SMALLINT (e.g., 2024)

| |— MEDIUMINT → Population MEDIUMINT (e.g., 500000)

| |— INT → Student_ID INT (e.g., 101)

| |— BIGINT → Phone_Number BIGINT (e.g., 9876543210)

| |— FLOAT → Height FLOAT (e.g., 5.9)

| |— DOUBLE/NUMERIC → Salary DOUBLE (e.g., 55000.75)

| |— BIT → Is_Active BIT (e.g., 1)

|

|— Date and Time

| |— DATE → DOB DATE (e.g., '2003-05-14')

| |— TIME → Login_Time TIME (e.g., '09:30:00')

| |— DATETIME → Created_At DATETIME (e.g., '2026-03-02 10:15:00')

| |— TIMESTAMP → Updated_At TIMESTAMP (auto current time)

| |— YEAR → Passing_Year YEAR (e.g., 2026)

|

|— String

- | | — CHAR → Gender CHAR (1) (e.g., 'M')
- | | — VARCHAR → Name VARCHAR (50) (e.g., 'Karthikeya')
- | | — TEXT → Address TEXT
- | | — BINARY → Binary_Code BINARY (4)
- | | — VARBINARY → File_Data VARBINARY (255)
- | | — BLOB → Image BLOB
- |
- | — Special Data Types
 - | — BOOLEAN → Is_Student BOOLEAN (TRUE/FALSE)
 - | — JSON → Details JSON (e.g., {"marks":85,"grade":"A"})

Signed int – positive and negative

How to create a database

Create database database_name;

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

school

- Tables
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sys

Administration

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Information

Schema: school

Object Info

Session

Query 1

Limit to 1000 rows

Jump to

```
1  -- CREATE DATABASE
2  • CREATE DATABASE SCHOOL;
3
```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help

Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	09:13:05	CREATE DATABASE SCHOOL	1 row(s) affected	0.094 sec

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Schema: school

Query 1

```
1 -- CREATE Database
2 CREATE DATABASE SCHOOL;
3 -- selecting database
4 USE SCHOOL;
5 -- creating table students
6 CREATE TABLE STUDENTS(
7     ID INT,
8     NAME VARCHAR(30),
9     CLASS TINYINT
10 );
11
```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	09:13:05	CREATE DATABASE SCHOOL	1 row(s) affected	0.094 sec
✓ 2	09:16:55	USE SCHOOL	0 row(s) affected	0.015 sec
✓ 3	09:18:29	CREATE TABLE STUDENTS(ID INT, NAME ...	0 row(s) affected	0.125 sec

Data insertion:

1. Inserting all column values (in table order):

```
INSERT INTO table_name VALUES (value1, value2, value3);
```

Example:

```
INSERT INTO students VALUES (101, 'Ram', 9);
```

2. Inserting specific columns:

```
INSERT INTO table_name (column1, column2) VALUES (value1, value2);
```

Example:

```
INSERT INTO students (ID, Name) VALUES (101, 'Ram');
```

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Query 1

```
1 -- CREATE Database
2 • CREATE DATABASE SCHOOL;
3 -- selecting database
4 • USE SCHOOL;
5 -- creating table students
6 • CREATE TABLE STUDENTS(
7     ID INT,
8     NAME VARCHAR(30),
9     CLASS TINYINT
10 );
11 -- Inserting values into tables
12 • INSERT INTO STUDENTS VALUES(101, 'RAM', 9);
```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	09:13:05	CREATE DATABASE SCHOOL	1 row(s) affected	0.094 sec
✓ 2	09:16:55	USE SCHOOL	0 row(s) affected	0.015 sec
✓ 3	09:18:29	CREATE TABLE STUDENTS(ID INT, NAME ...	0 row(s) affected	0.125 sec
✓ 4	09:35:36	INSERT INTO STUDENTS VALUES(101, 'RAM', 9)	1 row(s) affected	0.031 sec

Error:

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' panel with a tree view showing 'school' and 'sys' databases. The main editor window shows a SQL query with the following content:

```
1  -- CREATE Database
2  • CREATE DATABASE SCHOOL;
3  -- selecting database
4  • USE SCHOOL;
5  -- creating table students
6  • CREATE TABLE STUDENTS(
7      ID INT,
8      NAME VARCHAR(30),
9      CLASS TINYINT
10 );
11 -- Inserting values into tables
12 • INSERT INTO STUDENTS VALUES(101, 'RAM', 9);
13 • INSERT INTO STUDENTS VALUES(101, 'RAM');
14
15
```

The right sidebar shows the 'SQLAdditions' panel with a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."

The bottom panel shows the 'Output' window with the 'Action Output' tab selected. It displays a table of execution results:

#	Time	Action	Message	Duration / Fetch
1	09:13:05	CREATE DATABASE SCHOOL	1 row(s) affected	0.094 sec
2	09:16:55	USE SCHOOL	0 row(s) affected	0.015 sec
3	09:18:29	CREATE TABLE STUDENTS(ID INT, NAME ...	0 row(s) affected	0.125 sec
4	09:35:36	INSERT INTO STUDENTS VALUES(101, 'RAM', 9)	1 row(s) affected	0.031 sec
5	09:36:57	CREATE DATABASE SCHOOL	Error Code: 1007. Can't create database 'school'; dat...	0.000 sec

The error message in the last row indicates that the database 'school' already exists, which is why the 'CREATE DATABASE' statement failed.

MySQL Workbench

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Query 1

```

11 -- Inserting values into tables
12 INSERT INTO STUDENTS VALUES(101, 'RAM', 9);
13 INSERT INTO STUDENTS VALUES(102, 'RAVI', 10);
14 -- Inserting values by specifying the column names
15 INSERT INTO STUDENTS(NAME, ID, CLASS)
16 VALUES('SIVA', 101, 9), ('SRINU', 103, 9);
17 -- SELECTING DATA FROM TABLE
18 SELECT * FROM STUDENTS;
19

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

ID	NAME	CLASS
101	RAM	9
101	SIVA	9
103	SRINU	9
102	RAVI	10

Administration Schemas

Information

Schema: school

STUDENTS 2

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
6	09:40:38	INSERT INTO STUDENTS(NAME, ID, CLASS) V...	2 row(s) affected Records: 2 Duplicates: 0 Wami...	0.015 sec
7	09:42:46	INSERT INTO STUDENTS VALUES(102, 'RAVI',...	1 row(s) affected	0.016 sec
8	09:44:05	SELECT * FROM STUDENTS LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
9	09:45:13	EXPLAIN SELECT * FROM STUDENTS	OK	0.000 sec
10	09:45:13	EXPLAIN FORMAT=JSON SELECT * FROM ST...	OK	0.000 sec
11	09:46:12	SELECT * FROM STUDENTS LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

-- ADD NEW COLUMN TO STUDENT TABLE

-- ALTERING TABLE

-- SYNTAX: ALTER TABLE TABLE_NAME ADD COLUMN COLUMN_NAME DATATYPE

The screenshot shows the MySQL Workbench interface. The main editor window displays a SQL script with the following content:

```
-- ALTERING TABLE
-- SYNTAX: ALTER TABLE TABLE_NAME ADD COLUMN COLUMN_NAME DATATYPE
ALTER TABLE STUDENTS ADD COLUMN AGE TINYINT;
SELECT *FROM STUDENTS;

-- DELETE COLUMN
-- DELETE CLASS COLUMN
-- SYNTAX ALTER TABLE TABLE_NAME DROP COLUMN COLUMN_NAME;
```

Below the editor, the 'Result Grid' is visible, showing the results of the 'SELECT *FROM STUDENTS;' query. The grid has columns ID, NAME, CLASS, and AGE. The data is as follows:

ID	NAME	CLASS	AGE
101	RAM	9	NULL
101	SIVA	9	NULL
103	SRINU	9	NULL
102	RAVI	10	NULL

On the right side, a 'SQLAdditions' panel displays a message: 'Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.'

At the bottom, the 'Output' panel shows the execution log for the 'STUDENTS' table. The log includes the following entries:

#	Time	Action	Message	Duration / Fetch
10	09:45:13	EXPLAIN FORMAT=JSON SELECT * FROM STUDEN...	OK	0.000 sec
11	09:46:12	SELECT * FROM STUDENTS LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
12	09:53:46	SELECT *FROM STUDENTS LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
13	09:53:53	SELECT *FROM STUDENTS LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
14	09:54:34	ALTER TABLE STUDENTS ADD COLUMN AGE TINYL...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
15	09:54:43	SELECT *FROM STUDENTS LIMIT 0, 1000	4 row(s) returned	0.016 sec / 0.000 sec

-- DELETE COLUMN

-- SYNTAX ALTER TABLE TABLE_NAME DROP COLUMN COLUMN_NAME;

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Query 1

Limit to 1000 rows

```

21 -- SYNTAX: ALTER TABLE TABLE_NAME ADD COLUMN COLUMN_NAME DATATYPE
22 ALTER TABLE STUDENTS ADD COLUMN AGE TINYINT;
23 SELECT *FROM STUDENTS;
24 -- DELETE COLUMN
25 -- DELETE CLASS COLUMN
26 -- SYNTAX ALTER TABLE TABLE_NAME DROP COLUMN COLUMN_NAME;
27 ALTER TABLE STUDENTS DROP COLUMN CLASS;
28 SELECT *FROM STUDENTS;
29

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

ID	NAME	AGE
101	RAM	NULL
101	SIVA	NULL
103	SRINU	NULL
102	RAVI	NULL

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Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
12	09:53:46	SELECT *FROM STUDENTS LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
13	09:53:53	SELECT *FROM STUDENTS LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
14	09:54:34	ALTER TABLE STUDENTS ADD COLUMN AGE TINY...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
15	09:54:43	SELECT *FROM STUDENTS LIMIT 0, 1000	4 row(s) returned	0.016 sec / 0.000 sec
16	09:55:48	ALTER TABLE STUDENTS DROP COLUMN CLASS	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
17	09:56:15	SELECT *FROM STUDENTS LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

-- DESCRIBE TABLE

-- SYNTAX: DESCRIBE TABLE_NAME

MySQL Workbench

Local instance MySQL80 x

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Query 1 x

Limit to 1000 rows

```

32 -- DELETE CLASS COLUMN
33 -- SYNTAX ALTER TABLE TABLE_NAME DROP COLUMN COLUMN_NAME;
34 ALTER TABLE STUDENTS DROP COLUMN CLASS;
35 SELECT *FROM STUDENTS;
36
37 -- DESCRIBE TABLE
38 -- SYNTAX : DESCRIBE TABLE_NAME
39 DESCRIBE STUDENTS;
40

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

Field	Type	Null	Key	Default	Extra
ID	int	YES		NULL	
NAME	varchar(30)	YES		NULL	
AGE	tinyint	YES		NULL	

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Schema: school

Result 7 x

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Output

Action Output

#	Time	Action	Message	Duration / Fetch
13	09:53:53	SELECT *FROM STUDENTS LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
14	09:54:34	ALTER TABLE STUDENTS ADD COLUMN AGE TINYI...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
15	09:54:43	SELECT *FROM STUDENTS LIMIT 0, 1000	4 row(s) returned	0.016 sec / 0.000 sec
16	09:55:48	ALTER TABLE STUDENTS DROP COLUMN CLASS	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
17	09:56:15	SELECT *FROM STUDENTS LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
18	09:58:22	DESCRIBE STUDENTS	3 row(s) returned	0.016 sec / 0.000 sec

Object Info Session

CHANGE COLUMN DATA TYPE

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Query 1

Limit to 1000 rows

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```

35 • SELECT *FROM STUDENTS;
36
37 -- DESCRIBE TABLE
38 -- SYNTAX : DESCRIBE TABLE_NAME
39 • DESCRIBE STUDENTS;
40
41 -- CHANGE AGE COLUMN DATATYPE
42 • ALTER TABLE STUDENTS MODIFY COLUMN AGE INT;
43

```

Result Grid

Filter Rows:

Exports:

Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
ID	int	YES		NULL	
NAME	varchar(30)	YES		NULL	
AGE	int	YES		NULL	

Form Editor

Field Types

Administration

Information

No object selected

Result 9

Read Only

Context Help

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Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 17	09:56:15	SELECT *FROM STUDENTS LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
✓ 18	09:58:22	DESCRIBE STUDENTS	3 row(s) returned	0.016 sec / 0.000 sec
✓ 19	10:03:52	ALTER TABLE STUDENTS MODIFY COLUMN AGE INT	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.094 sec
✓ 20	10:04:18	ALTER TABLE STUDENTS MODIFY COLUMN AGE INT	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
✓ 21	10:04:27	SELECT *FROM STUDENTS LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
✓ 22	10:04:34	DESCRIBE STUDENTS	3 row(s) returned	0.000 sec / 0.000 sec

Object Info

Session

CONSTRAINS:

- 1) NOT NULL
- 2) UNIQUE
- 3) PRIMARY KEY
- 4) FORIEN KEY

- 5) AUTO_INCREMENT
- 6) DEFAULT
- 7) CHECK

NOT NULL – Ensures a column cannot store NULL (empty) values.

UNIQUE – Ensures all values in a column are different (no duplicates allowed).

PRIMARY KEY – Uniquely identifies each record in a table (combination of NOT NULL + UNIQUE).

FOREIGN KEY – Links from one table to another by referencing the primary key of another table.

AUTO_INCREMENT – Automatically generates sequential numeric values for a column.

DEFAULT – Assigns a default value to a column if no value is provided during insertion.

CHECK – Ensures that column values satisfy a specified condition.



